

Rizq Rayyan Ardiyanta Mohd Syahrir Kartayasa Putra

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EDUCATION

Nanyang Polytechnic

Diploma in Game Development & Technology

Expected Graduation: Apr 2027

Relevant Modules: Object-Oriented Programming, Data Structures & Algorithms, Game Engine Application, Web Development, Computer Graphics, Mobile App Development, Shader Optimisation, Multiplayer Game Programming, Cloud Computing

SKILLS

Programming Languages: C++, C#, JavaScript, HTML, CSS, MySQL, XML

Engines & Tools: Unity, Android Studio, OpenGL

Architecture: Finite State Machines (FSM), A* Pathfinding, Custom AI

Version Control & Cloud: Git, GitHub, AWS, Oracle Cloud (OCI)

PROJECT EXPERIENCE

Self-Hosted Game Server Infrastructure

Solo Developer | Sep 2026 – Present

- Deployed and administer two heavily-modded Minecraft servers (300–400+ mods each) on Oracle Cloud Ampere (ARM) Linux instances, tuning JVM heap and garbage-collection settings to run both stably within the free tier's shared resource allocation for 12 concurrent players (4:8 split).
- Built a custom Discord music bot using Node.js, handling audio streaming and playback commands via the Discord API.

“Chromatic Decay” — Year 3 Specialisation Project

Solo Developer | Apr 2026 – Aug 2026

- Solo-developed a first-person 3D momentum platformer in Unity (C#) over a 17-week production cycle as a capstone Specialisation Project, owning design, programming, art, and audio end-to-end.
- Engineered a fully diegetic, no-HUD feedback system: a shader-driven screen distortion and real-time audio low-pass filter that punish standing still, replacing conventional health bars or meters entirely.
- Built an advanced momentum-based movement kit (Dash, Super Jump, Wavedash, Hyperdash, Wallrun) with a 0.2s input-buffer system and custom physics overrides, including dynamic ground friction and vector-projected wall contact, to preserve player momentum.
- Ran a Beta playtest with 11 external testers and shipped a post-Beta patch resolving 7 identified issues, adding accessibility sliders for brightness, effect intensity, and audio.

AR Halal Ingredient List Checker

Solo Developer | May 2026

- Built an Augmented Reality (AR) mobile app that scans and verifies halal ingredients against a live database.
- Programmed real-time scanning and database retrieval using C# and Unity.

"Celeste" Mobile Clone

Gameplay & Systems Programmer | Feb 2026

- Programmed a fully playable "Celeste" mobile clone for Android.
- Reverse-engineered platforming physics in JavaScript (Android Studio).
- Built a custom XML level reader for dynamic stage loading.

Personal Portfolio Website

Solo Developer | Sep 2025

- Designed and deployed a responsive front-end using HTML, CSS, and JavaScript.
- Hosted on GitHub Pages to showcase development projects.

"VVVVVV" Clone

Solo Developer | Aug 2025

- Developed a clone by adapting to a custom C++ OpenGL framework.
- Engineered enemy behaviors and core loops using FSMs.
- Replicated complex physics and collision detection systems.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Online Gaming Community (Destiny 2 Clan)

Moderator & Team Coordinator | 2019 - Present

- Managed a large-scale active online community of over 150 members and regularly organized events.
- Coordinated 6-player teams over 400 instances through highly complex cooperative challenges.

Edgefield Secondary School Prefectorial Board

Senior Prefect (Internal Affairs & External Affairs) | 2020 - 2024

- Managed student affairs and served as a core organizer for the annual Open House.
- Coordinated student deployments and maintained school discipline.